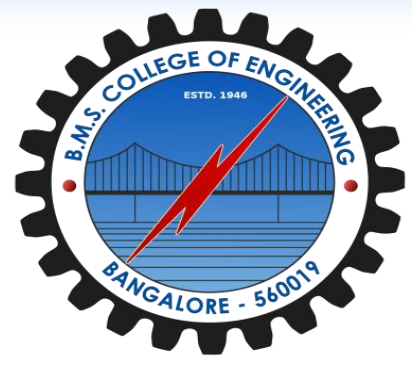




Solar Panel Performance Monitoring and Predictive Maintenance System



INTRODUCTION

- Solar panels gradually lose efficiency due to dust accumulation, overheating, environmental variations, and electrical faults. Continuous monitoring helps detect these issues early, but manual inspection is time-consuming and inaccurate.
- This project develops a **low-cost Solar Panel Performance Monitoring and Predictive Maintenance System** using real-time sensor data and machine learning.
- A prototype setup with a solar panel, LM35 temperature sensor, DHT11 humidity sensor, ACS712 current sensor, Dust Sensor GP2Y1010AU0F, Irradiance and a DC voltage module is interfaced with Arduino. The captured parameters—voltage, current, temperature, humidity, dust level and power—are logged into Excel and used to build a predictive model capable of identifying conditions such as normal operation, dust accumulation, overheating, and early-stage faults.
- The system provides a simple, scalable, and accurate approach to maintaining solar panels, improving performance and extending panel life.

OBJECTIVES

- To Provide Predictive Maintenance Alerts to Minimize Downtime.
- To Detect Performance Degradation Using Data Analytics.
- To Monitor Real-Time Performance of Solar Panels.

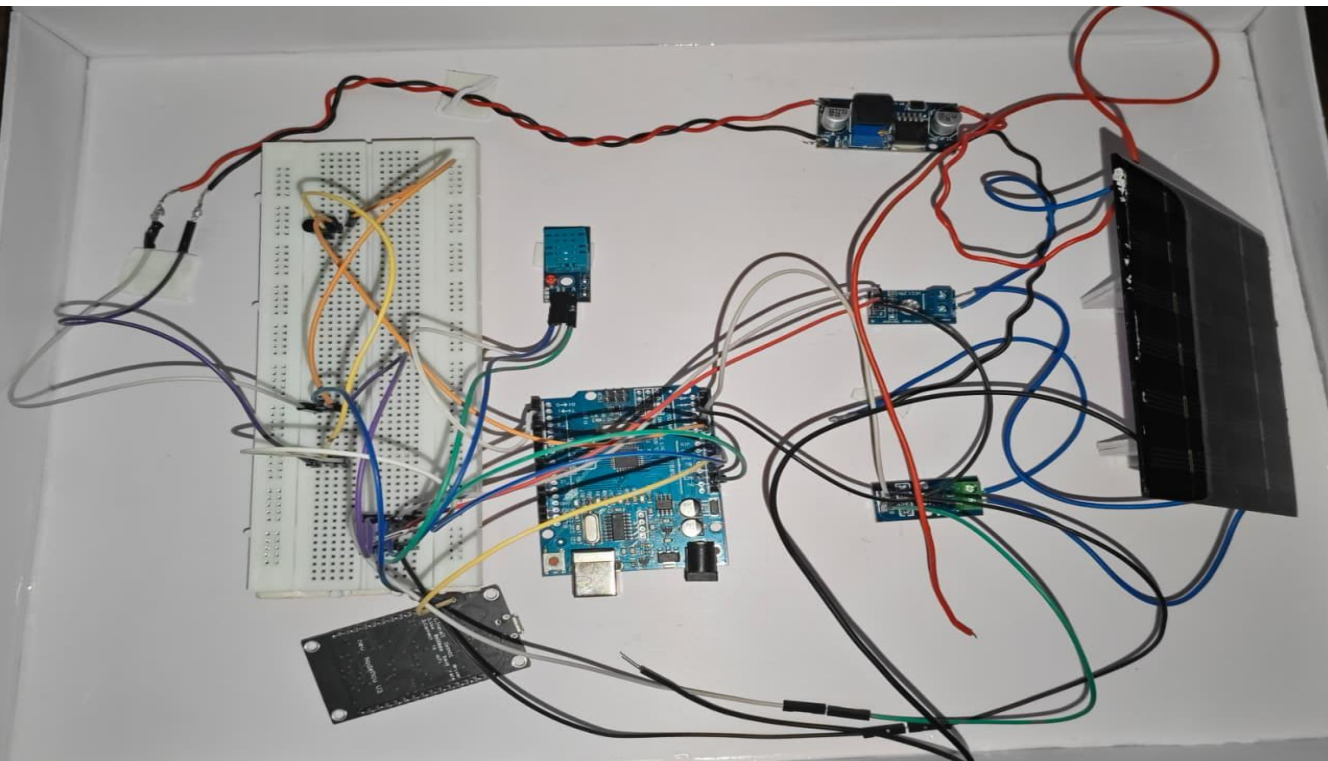
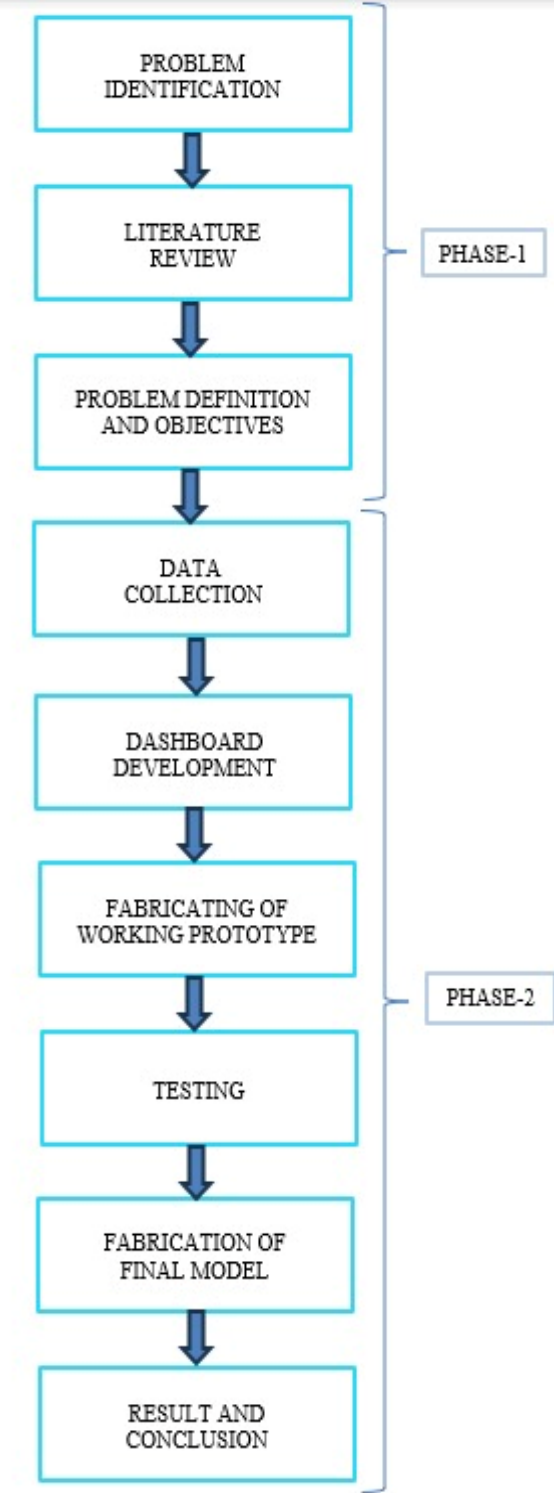


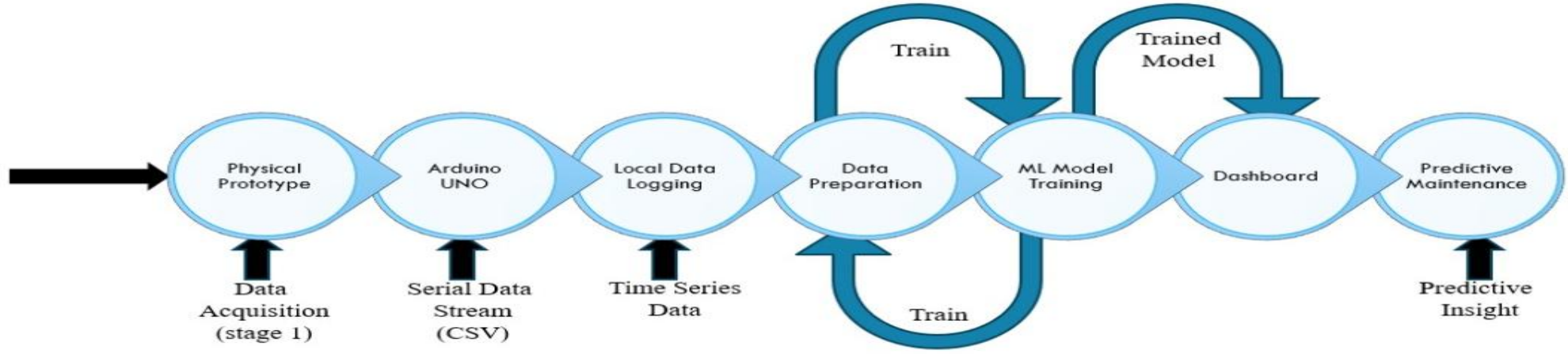
Fig. Working Prototype

METHODOLOGY



- Sensor Integration:** All sensors were connected to the Arduino Uno.
- Data Acquisition:** Arduino continuously reads voltage, current, panel temperature, air temperature, and humidity values every 2 seconds.
- Data Logging:** Values are formatted into CSV and sent to MS Excel via PLX-DAQ, which automatically logs readings with timestamps.
- Dataset:** Real-time readings combined with simulated data to cover multiple panel conditions.
- Development:** React frontend, Python backend, MongoDB, and Random Forest for predictions.
- Integration:** Dashboard uploads data, shows trends, performance metrics, and health predictions.

SYSTEM ARCHITECTURE:



RESULTS

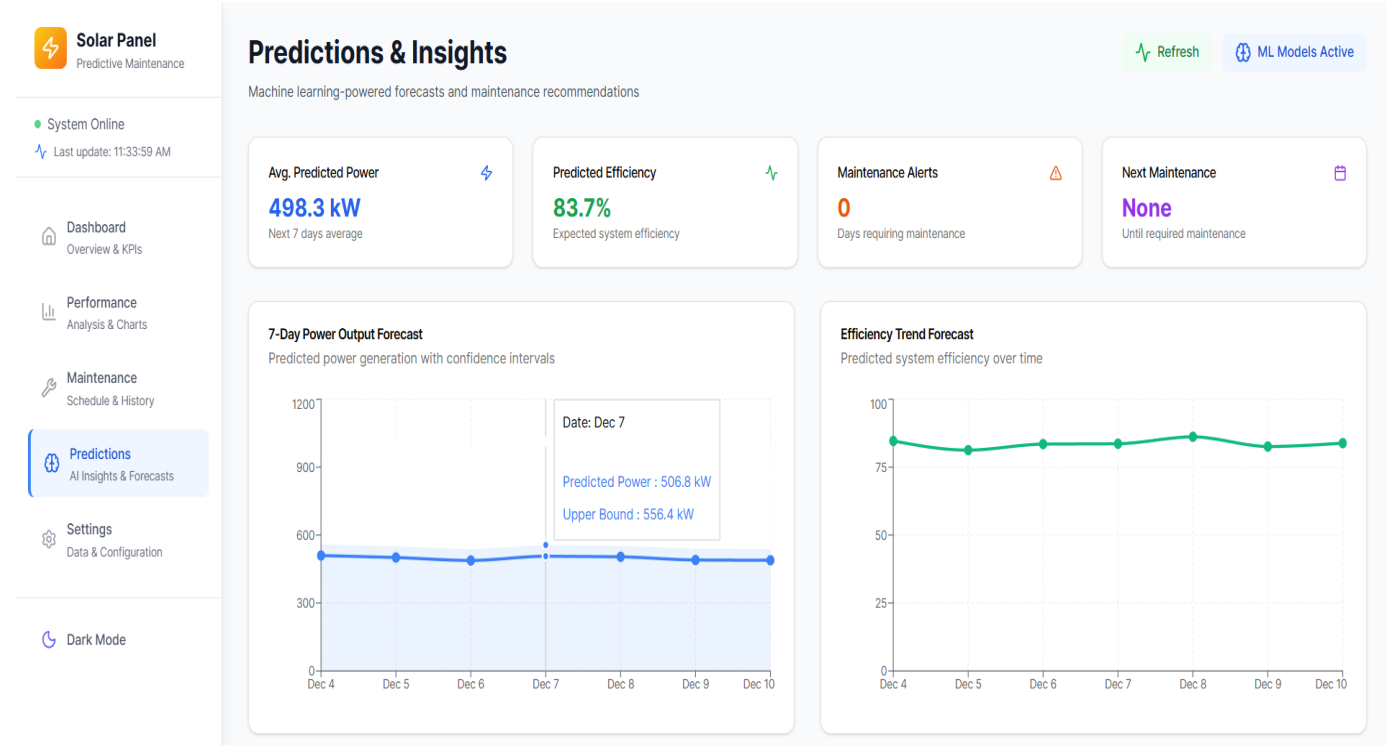


Fig. Predictions & Insights

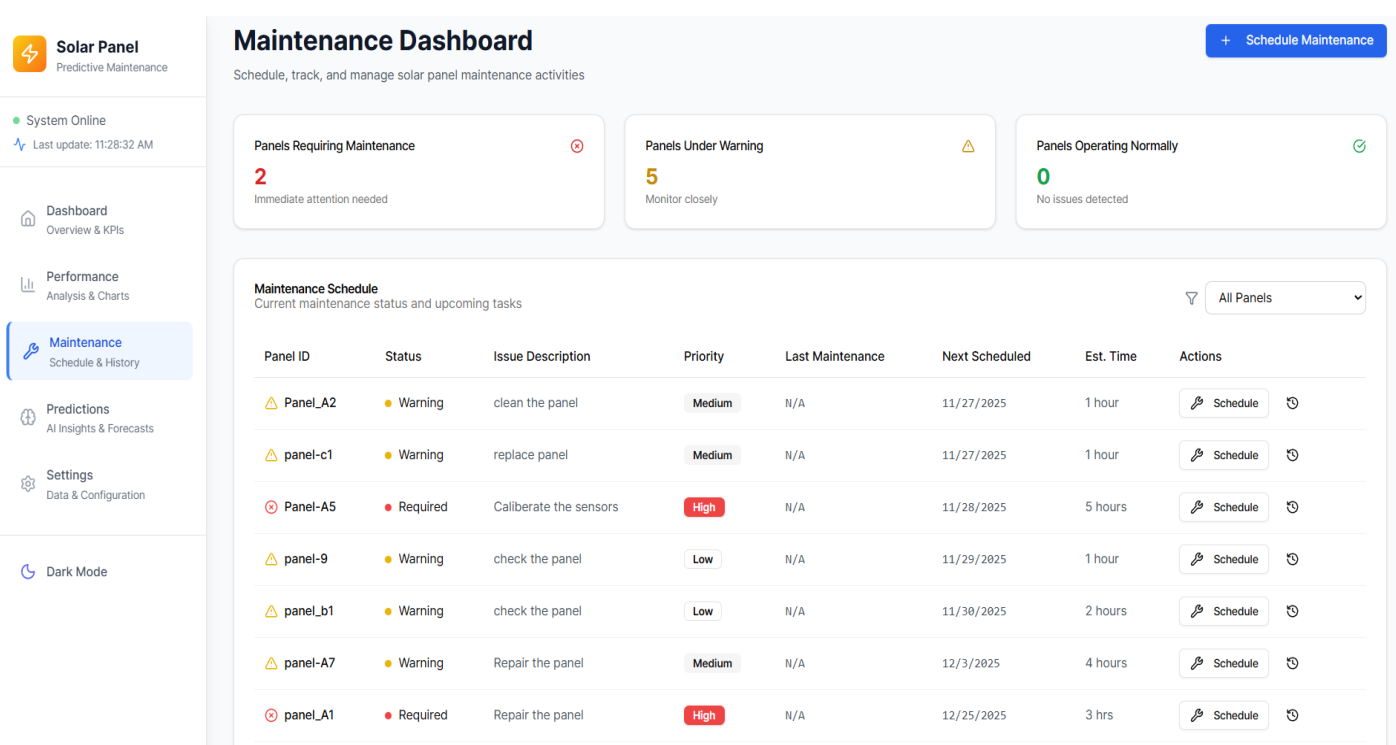


Fig. Maintenance Dashboard

MATERIALS & METHODS

Materials Used

- Solar Panel (9V, 100mA): Primary source for voltage and current generation.
- Arduino Uno: Main microcontroller used for data acquisition.
- LM35 Sensor: Measures panel surface temperature.
- DHT11 Sensor: Records ambient air temperature and humidity.
- ACS712 Current Sensor: Measures current output from the panel.
- DC Voltage Sensor Module: Monitors real-time panel voltage.
- Breadboard & Jumper Wires: Used for all prototype circuit connections.
- PLX-DAQ / MS Excel: Used for real-time data logging and dataset creation.

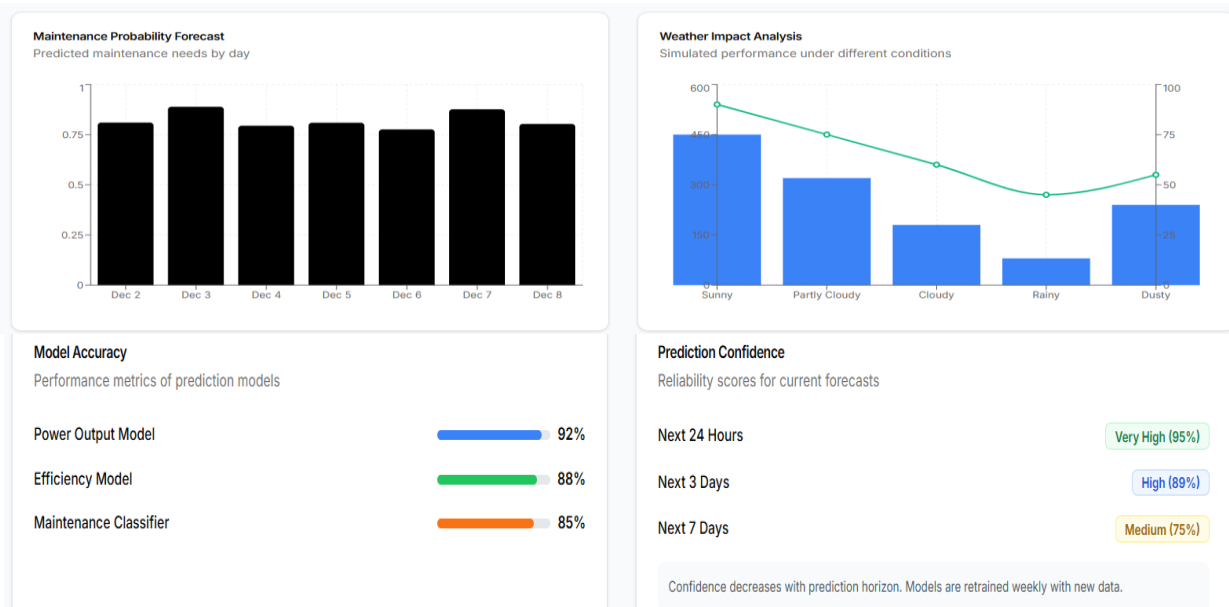


Fig. Predictive model performance

The prototype captured real-time solar panel parameters and logged clean datasets for analysis, enabling accurate Random Forest predictions for efficiency, power, and 7-day maintenance forecasting (85–95% accuracy). A live dashboard provides KPI monitoring, trend charts, alerts, and AI-driven maintenance recommendations that reduce operational costs.

CONCLUSION

The developed system successfully combines real-time sensor monitoring with machine learning to accurately assess solar panel performance and predict maintenance needs. The prototype and dashboard demonstrated reliable detection of efficiency drops, dust accumulation, and overheating, enabling proactive maintenance and improved energy output. This low-cost, scalable solution shows strong potential for practical solar panel health monitoring applications.

Project By

C P SAGAR - 1BM22IM013

IMAMSAB - 1BM22IM021

SHIVKUMAR M - 1BM22IM049

Under the guidance of

Prof. Disha M Nayak
(Assistant Professor)

SDG 7: Affordable and Clean Energy — this project supports reliable, efficient, and sustainable solar energy through smart monitoring and predictive maintenance.

SDG 11: Promotes sustainable communities by enhancing reliable solar energy use in urban systems.

SDG 13: Contributes to climate action by reducing emissions through optimized renewable energy performance.